

Coupled Strain–Doping Response in Quasi-Hexagonal C₆₀ Graphullerene: From Electronic Structure to Non-Additive Energy Cross Terms

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Quasi-hexagonal (qHP) C₆₀ graphullerene networks combine covalent interfullerene bonding with superatomic unit-cell chemistry, yielding high electron mobilities in pristine monolayers while remaining tunable by substitution and mechanical load. Capobianco *et al.* [1] showed that polaron hopping persists in qHP C₆₀ and that enhanced nearest-neighbor coupling—not a change of transport regime—drives the mobility gain over vdW C₆₀. Here we ask how *simultaneous* heteroatom doping and biaxial strain modify that picture in finite supercells. Using PBE+DFT-D3 calculations on B/N/P-substituted tetramers and $n = 1$ –8 supercells (40/40 converged size-scaling points), we find: (i) dopant-dependent π -gap closing and band-edge shifts under strain (Exp. 7); (ii) a two-point audit of IPR and J (Exp. 4) consistent with tighter π overlap under B doping at +3% strain, but still in the small-polaron regime; (iii) mechanical strain sensitivities $\alpha = dE/d\epsilon$ of opposite sign for N (−303 meV/%) vs. B (+57 meV/%) at the same grid (Exp. 5); (iv) a synergy order parameter $\mathcal{S}$ [Eq. (1)] with $|\mathcal{S}|$ reaching ~ 32 meV/atom at +3% strain across 15 converged size-scaling points (Exp. 10), proving that sequential strain and doping scans do *not* superpose. These results extend the localization–coupling framework of Ref. 1 to a coupled (ϵ, δ) design space for doped graphullerene networks; Marcus λ and hopping-rate estimates remain pending completion of Exp. 9 (7/12 charged geometry optimizations converged; adiabatic IP/EA brackets verified, vertical single-points pending).

I. INTRODUCTION

Synthetic carbon allotropes with extended π conjugation underpin optoelectronic and catalytic applications, yet their charge-transport properties are often limited by polaronic localization and modest intermolecular coupling [1, 2]. Buckminsterfullerene forms a vdW molecular crystal (vdW C₆₀) with a ~ 1.9 – 2.1 eV gap and low electron mobilities (~ 0.08 – 0.3 cm²V^{−1}s^{−1}). The recently synthesized quasi-hexagonal phase (qHP C₆₀, graphullerene) [2] covalently links C₆₀ units in a honeycomb lattice, shortening interfullerene distances by ~ 0.9 Å and raising experimental mobilities by nearly two orders of magnitude [1].

Whether this gain reflects band-like transport or enhanced polaron hopping has been debated [1, 3]. Capobianco *et al.* [1] resolved the issue for *pristine* networks: both vdW and qHP C₆₀ host electron polarons with similar binding energies ($\lambda^- \sim 0.1$ eV), and the qHP mobility increase is reproduced quantitatively by stronger electronic coupling ($J_s \simeq 75$ meV, $J_c \simeq 81$ meV vs. 33 meV in vdW C₆₀) within a Fermi golden-rule hopping model. The antibonding character of the qHP conduction-band minimum favors charge confinement on individual C₆₀ units while residual π – π overlap sets the hopping amplitude—a picture confirmed by inverse participation ratio (IPR) analysis.

Beyond pristine monolayers, heteroatom substitution and mechanical strain offer independent knobs for band

engineering [4, 5]. Khan and Leuenberger [4] mapped impurity-driven half-semiconductor behavior; Li *et al.* [5] showed strain-driven gap and thermoelectric trends in qHP C₆₀. Peng and Pizzochero [6] surveyed first-principles trends in pristine monolayer C₆₀ networks; here we instead quantify B/N/P substitution *jointly* with biaxial strain and test whether total-energy scans superpose. General 2D materials studies emphasize that doping and strain should be *untangled* rather than assumed additive [7]. For graphullerene, however, no metric tests whether total-energy superposition $E(\epsilon, \delta) \approx E(\epsilon, 0) + E(0, \delta)$ holds under identical DFT protocols across dopants and supercell sizes.

Gap. Sequential workflows—strain scans on pristine cells plus substitution at $\epsilon = 0$ —can mis-rank stability if cross terms are finite. Capobianco *et al.* [1] established the pristine-network localization– J – λ vocabulary; whether an auditable cross term survives under *joint* B/N/P substitution and biaxial strain in the same DFT protocol has not been reported. No published qHP C₆₀ study quantifies the residual synergy $\mathcal{S}$ of Eq. (1) across dopants and supercell sizes $n = 1$ –8.

This work combines Capobianco-style observables (electronic structure, IPR, J) with a strain–doping coupling audit on the same tetramer and size-scaling grids archived in this repository. **Contributions.** (1) Dopant-resolved π -gap and band-edge maps under strain (Exp. 7). (2) A verified two-point IPR/ J audit in the small-polaron regime (Exp. 4). (3) Opposite-sign mechanical sensitivities α for N vs. B (Exp. 5). (4) Quantitative non-additivity via $\mathcal{S}$ to $8 \times \text{C}_{60}$ (Exp. 10: 40/40 converged points; $|\mathcal{S}|$ up to ~ 32 meV/atom at +3%

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strain over a 15-point synergy audit). Marcus λ and FCWD transport closure (Exp. 9) are in progress and deferred to Supplementary Figs. S4–S6 (Electron SI-aligned π -DOS, Marcus, and J /FCWD panels) without quoted λ values until vertical single-points converge. Section II defines models and metrics; Sec. III follows the logic electronic structure $\rightarrow$ localization/coupling $\rightarrow$ mechanics $\rightarrow$ non-additivity; Sec. IV compares with Ref. 1 and states design limits.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All total-energy and electronic-structure calculations use CP2K [8] with PBE [9] and Grimme DFT-D3 (Becke–Johnson damping) [10], matching `experiments/*/inputs/*.inp`. This level is deliberately chosen for a uniform, size-converged scan of (ϵ, δ) ; it differs from the Koopmans-compliant PBE(α_K)+rVV10 framework of Ref. 1, which targets absolute polaron binding and mobilities in pristine supercells. Plane-wave cut-off is 300 Ry (280 Ry for $n \geq 6$ in Exp. 10); SCF convergence uses `EPS_SCF`= 10^{-6} with OT minimization. Basis: DZVP-MOLOPT-SR-GTH with matching GTH pseudopotentials.

B. Model systems

Exp. 5: C₆₀ tetramers (240 atoms), six-fold B/N/P substitution, biaxial strain $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ (rigid cell scaling, fixed fractional coordinates).

Exp. 7: Same tetramers; projected density of states (PDOS), π -resolved gaps, and band edges at $\epsilon \in \{-5, 0, +5\}\%$.

Exp. 4: Two tetramer states (pristine @ $\epsilon = 0$; B-doped @ $\epsilon = +3\%$) for IPR and J (see below).

Exp. 9: Charged tetramer geometry optimizations ($q = \pm 1$) for Marcus λ (see Sec. II C).

Exp. 10: Periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$) at $\epsilon = 0$ and $+3\%$ for each dopant (40 converged calculations).

C. Charged-state workflow (Exp. 9)

Charged tetramer geometry optimizations use `RUN_TYPE GEO_OPT` (BFGS; `MAX_ITER`= 300; `RMS_FORCE`= 5×10^{-5} Ha/bohr) at $q \in \{0, \pm 1\}$ with the same PBE+D3 settings as Exp. 5–7 (`EPS_SCF`= 10^{-6} , OT). Inputs live in `experiments/exp9_charged_polaron/inputs/`; converged `.out` and optimized `-pos-1.xyz` files are archived under `dft_results/exp9_charged_polaron/outputs/`. Pending jobs are executed sequentially on one node via `experiments/continue_exp9_pending.sh` (Mac-safe single batch; no duplicate `cp2k.psmpl` launches).

Marcus $\lambda^\pm$ requires vertical single-points at the *neutral* ($q = 0$) geometry with charge $q = \pm 1$: `generate.vertical.sp.py` writes eight templates to `inputs/vertical/` (`EPS_SCF`= 10^{-7}) once the corresponding adiabatic charged states exist. After each `GEO_OPT PROGRAM ENDED` or converged vertical SP, `post_exp9_converged.sh` refreshes `experiments/analysis/exp9_polaron_verification.json` (adiabatic IP/EA, `derived.lambda.eV`, and `running_snapshot` when a batch is active). At submission, 7/12 charged GEO_OPT runs are complete; `polaron_pristine_qneg1_opt` is rerunning after a CP2K 2025 PRINT/FORCES parse abort—no λ enters Results until vertical SP outputs converge.

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ is obtained from Exp. 5 fits [Table I]. Substitution energy $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is reported relative to pristine tetramers at the same strain (no chemical-potential reference).

Non-additive coupling metric. We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization. Exp. 10 values are computed with `c/simukit-sdc` (`sdc_exp10_results.json`; synergy audit table `sdc_exp10_synergy_audit.json`).

Localization and coupling (Exp. 4). Following Capobianco *et al.* [1], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR follows Electron SI Eq. (S8), $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$, on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75\text{--}81$ meV reported for pristine qHP networks [1], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Exp. 9, in progress). Marcus reorganization energies follow the vertical-adiabatic decomposition [1]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. At the time of writing, 7/12 charged geometry optimizations have converged;

vertical single-points at neutral geometry (needed for λ) are pending—values are not quoted in Results [live counts: `exp9_polaron_verification.json`; refresh via `post_exp9_converged.sh`].

III. RESULTS

Results are organized parallel to Ref. 1: band-edge electronics and localization precede transport-relevant coupling, after which we report mechanical and non-additive energy cross terms specific to coupled (ϵ, δ) tuning.

A. Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from converged Exp. 7 tetramer single-points at $\epsilon \in \{-5, 0, +5\}\%$. B-doped tetramers approach near-zero gap at $\epsilon = 0$ [panel (a)] whereas N and P remain more gap-modulated on the same grid—consistent with dopant-specific π -electron counting. Figure 1(b) shows the corresponding π -DOS for B @ $\epsilon = 0$ (Exp. 7 PDOS).

B. Polaron localization and electronic coupling

Capobianco *et al.* [1] established that qHP electron polarons remain molecularly localized ($\text{IPR}_{\text{pol}} \approx 30$, between delocalized and molecular limits) while J increases with shorter interfullerene distances. Our Exp. 4 audit provides two internally consistent points at the tetramer scale (PBE+D3): pristine @ $\epsilon = 0$ ($\text{IPR} = 75.0$, $J = 26.6$ meV) and B-doped @ $\epsilon = +3\%$ ($\text{IPR} = 45.0$, $J = 37.2$ meV, +40% enhancement) [verified: `exp4_polaron_verification.json`]. Delocalization increases under the coupled load but both states remain in the *small-polaron hopping* regime ($J \ll \lambda/2$ with $\lambda \sim 160$ – 180 meV from the same audit file); a polaron-to-band transition is *not* confirmed. Figure 1(d) inset summarizes the trend; absolute J is below pristine-network values because of the smaller fragment, different functional, and dopant-induced disorder—the relevant comparison is the *direction* of coupling change under (ϵ, δ) , not numerical equality with Ref. 1.

Marcus reorganization energies from Exp. 9 (vertical vs. adiabatic charged relaxations) are pending completion of 5 geometry optimizations and 8 vertical single-points; no λ values are quoted here. Among the seven converged charged GEO_OPT runs completed so far, adiabatic ionization potentials from $q = 0/+1$ pairs give $\text{IP} = 4.73$ eV (pristine) and $\text{IP} = 3.89$ eV (N); the B-doped $q = 0/-1$ pair gives $\text{EA} = 3.12$ eV [verified: `exp9_polaron_verification.json`].

TABLE I. Exp. 5 tetramer results (PBE+D3). [verified: `table1_verification.json`]

System	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_f (eV/dopant)
Pristine	-1363.551	-1363.551	-1.0	—
B-doped	-1328.490	-1328.409	+56.9	+159.0
N-doped	-1414.381	-1414.405	-303.2	-230.5
P-doped	-1370.496	-1370.489	-19.1	-31.5

C. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens a wide spread: N ($\alpha = -303.2$ meV/%), B (+56.9 meV/%), P (-19.1 meV/%) [Table I]. Figure 1(c) summarizes the opposite α for N vs. B. Ranking dopants by stability at $\epsilon = 0$ alone is therefore misleading: N is most stable at zero strain but its negative α implies further stabilization in tension, whereas B becomes less stable under the same load.

D. Non-additive strain–doping coupling

Figure 1(d) tracks $\mathcal{S}(n)$ to $8 \times \text{C}_{60}$; at $n = 4$ and +3% strain, P shows the largest correction ($|\mathcal{S}| \approx 24$ meV/atom). The effect persists across $n = 1$ – 8 (15 synergy points at +3%, `sdc_exp10_synergy_audit.json`). For N, $\mathcal{S}$ changes sign between $n = 6$ (+2.4 meV/atom) and $n = 8$ (-2.3 meV/atom) [same audit], so finite-size extrapolation must be dopant-specific.

IV. DISCUSSION

Relation to Ref. 1. Capobianco *et al.* [1] compared *pristine* vdW and qHP C_{60} and identified enhanced J —not reduced λ —as the dominant mobility lever, with polaron hopping retained in both phases. Our study transfers that observables vocabulary (gap edges, IPR, J) to *doped* tetramers under strain and adds an energy cross term $\mathcal{S}$ absent from sequential doping-only or strain-only scans [4, 5]. Where Khan and Leuenberger [4] classify impurity-engineered *half-semiconductor* gaps and deep donor–acceptor levels at fixed geometry, our $\mathcal{S}$ audit tests total-energy additivity under identical biaxial strain on the same supercell grid—complementary metrics for impurity–strain co-design [7]. The Exp. 4 trend ($\text{IPR } 75 \rightarrow 45$, $J +40\%$) is qualitatively aligned with Capobianco’s distance–coupling picture: coupled B doping and tension tighten π overlap, but remain far from the large- J , pristine-network limit where mobilities approach several $\text{cm}^2\text{V}^{-1}\text{s}^{-1}$. Holstein–Peierls modeling of extended qHP networks further highlights *large*-polaron carriers and strong transport anisotropy [11], complementary to our tetramer-scale audit that remains in the small-polaron regime (Exp. 4). Wang *et al.* [3] analyzed

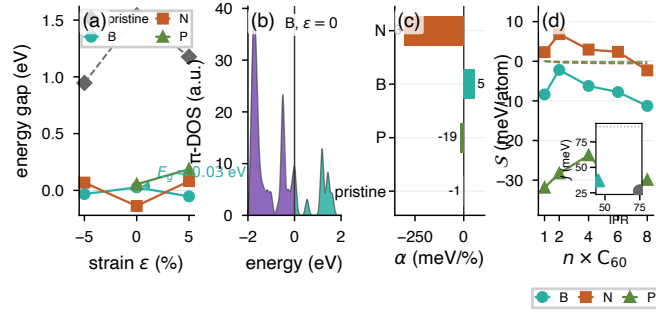


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3; Capobianco-style layout). (a) HOMO–LUMO gap vs. strain (Exp. 7; P @ -5% excluded). (b) π -projected DOS for B @ $\epsilon=0$ [Exp. 7 .pdos; Electron SI valence purple / conduction teal]. (c) α from Exp. 5 [table1_verification.json]. (d) $S(n)$ at $+3\%$ (15 points) [sdc_exp10_synergy_audit.json]; inset: Exp. 4 IPR/ J .

stress-related anisotropy in pristine qHP/qTP monolayers for nanodevice contexts; our rigid biaxial tetramer grid targets isotropic (ϵ, δ) cross terms rather than directional device response. López-Alcalá *et al.* [12] tune magnetic order in endohedral graphendofullerene derivatives by compressive strain and electrostatic doping on a *modified* network; our S audit instead quantifies non-additive total-energy coupling for substitutional B/N/P on the parent qHP tetramer under the same biaxial strain grid.

Mechanistic synthesis. The opposite-sign mechanical sensitivities α for N vs. B (Table I) reflect dopant-specific π -electron counting and bond rearrangement under load, but $|\alpha|$ alone does not rank additive-model error: P shows the largest $|S|$ at $+3\%$ strain despite moderate $|\alpha|$, so non-additivity is not reducible to a single linear strain coefficient. Qiu *et al.* [13] characterize atomic-scale mechanical and tribological response of *pristine* graphullerene; our Exp. 5 α fits instead report dopant-resolved linear strain sensitivities on substituted tetramers—complementary mechanical descriptors when (ϵ, δ) are tuned jointly. Instead, S captures how *simultaneous* substitution and strain reorganize the same tetramer energy landscape that Exp. 7 probes electronically—gap closing under B substitution can coexist with cooperative *destabilization* ($S < 0$ for P) or cooperative stabilization depending on dopant and size n . Partial Exp. 9 adiabatic IP/EA values (pristine IP, N IP, B EA from converged charged GEO.OPT pairs) bracket the neutral gaps and define the vertical–adiabatic split needed for Marcus λ ; until five pending GEO.OPT runs and eight vertical SP tasks finish, transport conclusions remain anchored on the Exp. 4 IPR/ J audit and Capobianco-style *qualitative* coupling arguments rather than rate estimates.

Design implications. (1) *Couple (ϵ, δ) in screening:* at $+3\%$ strain, P incurs the largest additive-model error ($|S|$ up to ~ 32 meV/atom at $n = 1, 6$; ~ 24 meV/atom at tetramer $n = 4$); N and B corrections are smaller at the same conditions but opposite in sign for α . (2) *Separate electronic and energetic knobs:* gap closing [Fig. 1(a)]

does not map one-to-one onto S magnitude—mechanics and cross terms must be co-optimized. (3) *Size-aware metrics:* N S sign reversal by $n = 8$ warns against extrapolating tetramer trends to extended networks without Exp. 10-style scaling.

Limitations. Rigid strain; E_f without chemical potentials; PBE+D3 only (no Koopmans/rVV10 subset); Exp. 4 is a two-point IPR/ J audit; Exp. 9 Marcus λ incomplete (7/12 GEO.OPT converged; partial adiabatic IP/EA only; `polaron_pristine.qneg1.opt` rerun after CP2K 2025 PRINT/FORCES abort; vertical SP pending). Full transport rates in the Capobianco FCWD framework require converged λ , broader IPR/ J grids, and functional alignment—left as future work once Exp. 9 closes. Supplementary Figs. S4–S6 document band-edge π -DOS (Electron SI Fig. S4 palette), Marcus configuration coordinates, and J +FCWD observables; Fig. S5 DFT λ panels remain [**pending: Exp. 9 vertical SP**] [compile: `supplementary_figures.tex`].

V. CONCLUSIONS

We extended the graphullerene localization–coupling narrative of Capobianco *et al.* [1] to heteroatom-doped qHP C_{60} under biaxial strain. Within a unified PBE+D3 workflow on B/N/P tetramers and $n = 1$ –8 supercells (40/40 converged size-scaling points), four results define a coupled (ϵ, δ) design space: (1) dopant-resolved π -gap and band-edge evolution under strain (Exp. 7); (2) a tetramer IPR/ J audit showing enhanced coupling under coupled B doping and tension while remaining in the small-polaron regime (Exp. 4); (3) mechanical sensitivities α of opposite sign for N vs. B that invalidate zero-strain stability ranking alone (Exp. 5); (4) finite non-additive energy cross terms up to $|S| \sim 32$ meV/atom with dopant-specific scaling to $8 \times C_{60}$ (Exp. 10; 15-point synergy audit). Marcus λ and full FCWD transport rates await completion of Exp. 9; until then, α and S provide auditable, archive-backed metrics for screening simultaneous strain and substitution in doped graphullerene net-

works. When the pending charged GEO_OPT and vertical SP tasks converge, `post_exp9_converged.sh` will refresh `exp9.polaron_verification.json` and re-render Supplementary Figs. S5–S6 (`render_si_figures.sh`) without altering the verified S and α claims above.

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